Midterm—Algebra*
Fall 2025

Algebra Team (2025)†

1. If A and B are $n \times n$ matrices, then:

$$(A+B)^2 = (A^2 + 2AB + B^2)$$

(a) true (b) false

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2. $x = 1$, $y = 2$ and $z = r$ is a solution of the linear system $\begin{cases} 2x + 3y - z = a \\ x - y + 2z = b \\ 4x + y - 2z = 8 \end{cases}$, where $a, b \in \mathbb{R}$. The value of a should be:

(a) 9 (b) -9 (c) 1 (d) -1 (e) none

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3. Consider the linear system $A\mathbf{x} = \mathbf{b}$. If A is an $n \times n$ matrix, and $\mathbf{b} \neq \mathbf{0}$, then the system cannot have:

(a) no solution (b) infinitely many solutions
(c) the trivial solution (d) none

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4. An elementary matrix E is used to perform the elementary row operation $-4r_3 + r_4 \to r_4$ on a 6×5 matrix using matrix multiplication. The entry in row 4 and column 3 of E^{-1} is:

(a) -4 (b) $1/4$ (c) $-1/4$ (d) 4 (e) none

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5. Consider the LU decomposition of the matrix $\begin{bmatrix} 3 & -6 & 3 \\ 9 & -17 & 16 \\ -6 & 14 & 4 \end{bmatrix}$. The entry in row 2 and column 3 of U is:

(a) 7 (b) -7 (c) 6 (d) -6 (e) none

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6. Let $A = B^T C^T D$, where $B = \begin{bmatrix} 1 & 2 & -3 \\ 4 & 1 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 3 & -4 \end{bmatrix}$, and $D = \begin{bmatrix} 2 & -1 & 7 \end{bmatrix}$. The entry in row 1 and column 3 of A is equal to:

(a) -91 (b) -65 (c) -78 (d) -52 (e) none

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7. If $x = 3$ is a root of the equation $2x^5 - 7x^4 - 3x^3 + a_2x^2 - 23x + 6 = 0$, then the value of a_2 is:

(a) -25 (b) 15 (c) 25 (d) -15 (e) none

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*This exam is not unified and uses multiple models (versions). All models are of an equivalent difficulty level.

†This material, which includes reproduced official exam questions and their solutions, is released purely for educational purposes (practice, review, and self-assessment).

8. Let $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 3 \\ 1 & -4 \end{bmatrix}$, and $XBA = C$. The entry in row 2 and column 1 of X is:

(a) -51 (b) -50 (c) -57 (d) -48 (e) none

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9. If $P(x) = x^5 - 3x^4 + 2x^3 + 6x^2 - 4x + 1$ is divided by $(x - 1 - i)$, the constant term of the quotient is:

(a) $1 + 8i$ (b) $4 + 4i$ (c) $1 - 8i$ (d) $10 - 7i$ (e) none

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10. If A, B and C are $n \times n$ nonsingular matrices, and $C = (A + B)^{-1}BA$, then C^{-1} is equal to:

(a) $(B^{-1}A + I)A^{-1}$ (b) $A^{-1}(AB^{-1} + I)$ (c) $A^{-1}(B^{-1}A + I)$
(d) $(AB^{-1} + I)A^{-1}$ (e) none

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11. The product of an $n \times n$ upper triangular and $n \times n$ lower triangular matrix is:

(a) diagonal (b) upper triangular (c) symmetric
(d) lower triangular (e) none

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12. Consider the linear system $\begin{cases} x + 2y - z + 5w = 3 \\ 2x + 3y + z + 8w = 5 \\ 3x + 4y + 3z + 11w = 7 \end{cases}$. If $z = t$, $w = s$ and $s, t \in \mathbb{R}$, then x is equal to:

(a) $1 - s - t$ (b) $1 + 6s + 7t$ (c) $1 - s - 5t$ (d) $1 + 7s + 5t$ (e) none

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13. The linear system $\begin{cases} x - 4y + 2z = 0 \\ x + 3y - 4z = 0 \\ rx + sy + tz = 0 \end{cases}$ will have nontrivial solutions if:

(a) $2r + s + t = 0$ (b) $r - 2s + 3t$ (c) $2r + 5s + 2t = 0$
(d) $10r + 6s + 7t = 0$ (e) none

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14. The linear system $\begin{cases} x + y + 2z = a \\ x + z = b \\ 2x + y + 3z = c \end{cases}$ has no solution when:

(a) $a = 1$, $b = 2$ and $c = 3$ (b) $a = 1$, $b = 1$ and $c = 4$ (c) $a = 2$, $b = 2$ and $c = 4$
(d) $a = 1$, $b = 1$ and $c = 2$ (e) none

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15. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 4 & 6 & -2 \\ 4 & 11 & 18 & 11 \\ -1 & 0 & 5 & 3 \\ 6 & 14 & 23 & 5 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

(a) -3 (b) -2 (c) 2 (d) 3 (e) none

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16. Let $A = \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 2 & 1 \\ 4 & 9 & -2 & 17 & 4 & 6 & 11 & 8 \\ 2 & 7 & 4 & 11 & 3 & 9 & 14 & 21 \end{bmatrix}$. The entry in row 1 and column 8 of the reduced row echelon form of A is equal to:

(a) 14 (b) -14 (c) -21 (d) 21 (e) none

17. (choices are missing) The linear system $\begin{cases} x+2y+5z=4 \\ 2x+5y+13z=8 \\ 4x+5y+12z=19 \\ 3x+9y+25z=k \end{cases}$ has a unique solution if k is equal to:

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18. (choices are missing) The roots of $P(x)=x^4+\frac{1}{2}x^2+3x+2=0$ may include:

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19. (choices are missing) Consider the polynomial equation $ax^3+x^2+bx+1=0$. If $x=r$ is a repeated root of multiplicity 3, and a, b, $r\in\mathbb{R}$, then:

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20. (choices are missing) The linear system $\begin{cases} 2x+y+3z-8w=4 \\ 3x+4y+z+2w=3 \\ 12x+11y+11z+aw=b \end{cases}$ is consistent when:

1 Answer key

1. If A and B are $n \times n$ matrices, then:

$$(A+B)^2 = (A^2 + 2AB + B^2)$$

Answer: (b) false;

$$\begin{aligned}(A+B)^2 &= (A+B)(A+B) \\ &= A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2\end{aligned}$$

hence generally, the statement is NOT true.

It is only true if A and B are matrices that commute (which, generally, is not the case).

2. $x = 1$, $y = 2$ and $z = r$ is a solution of the linear system $\begin{cases} 2x + 3y - z = a \\ x - y + 2z = b \\ 4x + y - 2z = 8 \end{cases}$, where $a, b \in \mathbb{R}$. The value of a should be:

Answer: (a) 9; we could substitute in the values for x, y, z and obtain the equations

$$\text{(eq. 1) } 2 + 6 - r = a \Rightarrow a + r = 8$$

$$\text{(eq. 2) } 1 - 2 + 2r = b \Rightarrow b - 2r = -1$$

$$\text{(eq. 3) } 4 + 2 - 2r = 8 \Rightarrow r = -1$$

solving for a yields $a = 9$.

3. Consider the linear system $A\mathbf{x} = \mathbf{b}$. If A is an $n \times n$ matrix, and $\mathbf{b} \neq \mathbf{0}$, then the system cannot have:

Answer: (c) the trivial solution; since the system is nonhomogeneous ($\mathbf{b} \neq \mathbf{0}$), then we have three possible types of solutions

i. a unique solution ($A^{-1}\mathbf{b}$, where $|A| \neq 0$)
ii. infinitely many solutions ($|A| = 0$)
iii. no solution ($|A| = 0$)

obviously, if there were a unique solution, then it could never be the trivial solution $\mathbf{x} = \mathbf{0}$ since this system is nonhomogeneous.

4. An elementary matrix E is used to perform the elementary row operation $-4r_3 + r_4 \to r_4$ on a 6×5 matrix using matrix multiplication. The entry in row 4 and column 3 of E^{-1} is:

Answer: (d) 4; the operation $-4r_3 + r_4 \to r_4$ is done using an elementary matrix with the entry -4 in the 4th row and the 3rd column, which means it lies off the main diagonal.

The corresponding inverse operation, then, is adding 4 times the third row to the fourth row, which is equivalent to the elementary matrix with the entry 4 in the 4th row and the 3rd column.

5. Consider the LU decomposition of the matrix $\begin{bmatrix} 3 & -6 & 3 \\ 9 & -17 & 16 \\ -6 & 14 & 4 \end{bmatrix}$. The entry in row 2 and column 3 of U is:

Answer: (a) 7; using Gaussian elimination

$$\xrightarrow{(1/3)r_1 \to r_1} \begin{bmatrix} 1 & -2 & 1 \\ 9 & -17 & 16 \\ -6 & 14 & 4 \end{bmatrix}$$

$$\xrightarrow[r_3+6r_1 \to r_3]{r_2-9r_1 \to r_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 7 \\ 0 & 2 & 10 \end{bmatrix}$$

obviously, the entry in the second row and third column of U is 7.

6. Let $A = B^T C^T D$, where $B = \begin{bmatrix} 1 & 2 & -3 \\ 4 & 1 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 3 & -4 \end{bmatrix}$, and $D = \begin{bmatrix} 2 & -1 & 7 \end{bmatrix}$. The entry in row 1 and column 3 of A is equal to:

Answer: (a) -91; writing A as the product of the matrices

$$A = \underbrace{\begin{bmatrix} 1 & 4 \\ 2 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -4 \end{bmatrix}}_{\text{produces a } 3 \times 1 \text{ matrix}} \begin{bmatrix} 2 & -1 & 7 \end{bmatrix}$$

then the entry in row 1 and column 3 of A is that produced from the multiplication of row 1 of $B^T C^T$ and column 3 of D.

Row 1 of $B^T C^T$ is $\begin{bmatrix} 1 \times 3 + 4 \times -4 \end{bmatrix} = \begin{bmatrix} -13 \end{bmatrix}$. Hence the entry is $-13 \times 7 = -91$.

7. If $x = 3$ is a root of the equation $2x^5 - 7x^4 - 3x^3 + a_2x^2 - 23x + 6 = 0$, then the value of a_2 is:

Answer: (c) 25; since $x = 3$ is a root of the polynomial equation

$$2(3)^5 - 7(3)^4 - 3(3)^3 + a_2(3)^2 - 23(3) + 6 = 0 \Rightarrow a_2 = 25$$

8. Let $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 3 \\ 1 & -4 \end{bmatrix}$, and $XBA = C$. The entry in row 2 and column 1 of X is:

Answer: (d) -48; solving for X

$$XBA = C \Rightarrow X = CA^{-1}B^{-1}$$

and by calculating the inverses of B and A

$$A^{-1} = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$$

then, multiplying the matrices

$$X = \begin{bmatrix} 41 & -112 \\ -48 & 131 \end{bmatrix}$$

9. If $P(x) = x^5 - 3x^4 + 2x^3 + 6x^2 - 4x + 1$ is divided by $(x - 1 - i)$, the constant term of the quotient is:

Answer: (b) $4 + 4i$; using synthetic division $(x = 1 + i)$

$1+i$	1	-3	2	6	-4	1
+		$1+i$	$-3-i$	$0-2i$	$8+4i$	
	1	$-2+i$	$-1-i$	$6-2i$	$\boxed{4+4i}$	

10. If A, B and C are $n \times n$ nonsingular matrices, and $C = (A + B)^{-1}BA$, then C^{-1} is equal to:

Answer: (c) $A^{-1}(B^{-1}A + I)$; taking the inverse

$$\begin{aligned} C^{-1} &= ((A + B)^{-1}BA)^{-1} \\ &= A^{-1}B^{-1}(A + B) = A^{-1}(B^{-1}A + I) \end{aligned}$$

11. The product of an $n \times n$ upper triangular and $n \times n$ lower triangular matrix is:

Answer: (e) none; generally, any L and U yield a completely random matrix with no specific type. Think of the LU decomposition of any matrix A.

12. Consider the linear system $\begin{cases} x+2y- & z+ & 5w=3 \\ 2x+3y+ & z+ & 8w=5 \\ 3x+4y+3z+11w=7 \end{cases}$. If $z=t$, $w=s$ and $s,\, t \in \mathbb{R}$, then x is equal to:

Answer: (c) $1-s-5t$; putting the system into augmented matrix form

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 3 \\ 2 & 3 & 1 & 8 & 5 \\ 3 & 4 & 3 & 11 & 7 \end{array}\right]$$

proceeding with row reduction

$$\xrightarrow[r_3-3r_1 \to r_3]{r_2-2r_1 \to r_2} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 3 \\ 0 & -1 & 3 & -2 & -1 \\ 0 & -2 & 6 & -4 & -2 \end{array}\right]$$

$$\xrightarrow{r_3-2r_2 \to r_3} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 5 & 3 \\ 0 & -1 & 3 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

yields the equations,

$$x+2y-z+5w=3 \Rightarrow x=3-(2y-z+5w)$$

$$-y+3z-2w=-1 \Rightarrow y=1+3z-2w$$

substituting $z=t$ and $w=s$ to get y and then x

$$y=1+3t-2s, \quad x=1-s-5t$$

13. The linear system $\begin{cases} x-4y+2z=0 \\ x+3y-4z=0 \\ rx+sy+tz=0 \end{cases}$ will have nontrivial solutions if:

Answer: (d) $10r+6s+7t=0$; putting the system into matrix equation form

$$\underbrace{\begin{bmatrix} 1 & -4 & 2 \\ 1 & 3 & -4 \\ r & s & t \end{bmatrix}}_{\text{coefficients}} \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

since the system is homogeneous, then it has one of two possible types of solutions

i. a unique solution (the trivial solution, $|A| \neq 0$)

ii. infinitely many solutions (the trivial solution **AND** the nontrivial solutions, $|A|=0$)

so, calculating the determinant of the coefficient matrix

$$\begin{vmatrix} 1 & -4 & 2 \\ 1 & 3 & -4 \\ r & s & t \end{vmatrix} = (16-6)r-(-4-2)s+(3+4)t$$

$$= \boxed{10r+6s+7t=0}$$

14. The linear system $\begin{cases} x+y+2z=a \\ x \quad\quad + \quad z=b \\ 2x+y+3z=c \end{cases}$ has no solution when:

Answer: (b) $a=1$, $b=1$ **and** $c=4$; putting the system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 1 & 0 & 1 & b \\ 2 & 1 & 3 & c \end{array}\right]$$

proceeding with Gaussian elimination

$$\xrightarrow[r_3-2r_1 \to r_3]{r_2-r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & -1 & -1 & c-2a \end{array}\right]$$

$$\xrightarrow{-r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & 1 & a-b \\ 0 & -1 & -1 & c-2a \end{array}\right]$$

$$\xrightarrow{r_3-r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & 1 & 1 & a-b \\ 0 & 0 & 0 & c-a-b \end{array}\right]$$

obviously, for the system to not have a solution

$$\#\text{ leading ones} < \#\text{ non-zero rows}$$

so, $c-a-b \neq 0 \Rightarrow c \neq a+b$, which is only satisfied by choice (b).

15. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 4 & 6 & -2 \\ 4 & 11 & 18 & 11 \\ -1 & 0 & 5 & 3 \\ 6 & 14 & 23 & 5 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

Answer: (d) 3; using the careful book-keeping method

$$\xrightarrow{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 4 & 11 & 18 & 11 \\ -1 & 0 & 5 & 3 \\ 6 & 14 & 23 & 5 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \\ & & & \end{bmatrix}$$

$$\xrightarrow[r_4-6r_1 \to r_4]{r_2-4r_1 \to r_2,\ r_3+r_1 \to r_3} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 3 & 6 & 15 \\ 0 & 2 & 8 & 2 \\ 0 & 2 & 5 & 11 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & & 0 & 0 \\ -1 & & & 0 \\ 6 & & & \end{bmatrix}$$

$$\xrightarrow{(1/3)r_2 \to r_2} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 2 & 8 & 2 \\ 0 & 2 & 5 & 11 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ -1 & & & 0 \\ 6 & & & \end{bmatrix}$$

$$\xrightarrow[r_4-2r_2 \to r_4]{r_3-2r_2 \to r_3} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 4 & -8 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ -1 & 2 & & 0 \\ 6 & 2 & & \end{bmatrix}$$

$$\xrightarrow{(1/4)r_3 \to r_3} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ -1 & 2 & 4 & 0 \\ 6 & 2 & & \end{bmatrix}$$

$$\xrightarrow{r_4-r_3 \to r_4} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 3 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ -1 & 2 & 4 & 0 \\ 6 & 2 & 1 & \end{bmatrix}$$

$$\xrightarrow{(1/3)r_4 \to r_4} \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ -1 & 2 & 4 & 0 \\ 6 & 2 & 1 & 3 \end{bmatrix}$$

then the entry is 3.

16. Let $A = \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 2 & 1 \\ 4 & 9 & -2 & 17 & 4 & 6 & 11 & 8 \\ 2 & 7 & 4 & 11 & 3 & 9 & 14 & 21 \end{bmatrix}$. The entry in row 1 and column 8 of the reduced row echelon form of A is equal to:

Answer: (b) -14; using Gauss-Jordan elimination

$$\xrightarrow[r_3-2r_1 \to r_3]{r_2-4r_1 \to r_2} \begin{bmatrix} 1 & 2 & -1 & 4 & 1 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 & 2 & 3 & 4 \\ 0 & 3 & 6 & 3 & 1 & 7 & 10 & 19 \end{bmatrix}$$

$$\xrightarrow[r_3-3r_2 \to r_3]{r_1-2r_2 \to r_1} \begin{bmatrix} 1 & 0 & -5 & 2 & 1 & -3 & -4 & -7 \\ 0 & 1 & 2 & 1 & 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 7 \end{bmatrix}$$

$$\xrightarrow{r_1-r_3 \to r_1} \begin{bmatrix} 1 & 0 & -5 & 2 & 0 & -4 & -5 & -14 \\ 0 & 1 & 2 & 1 & 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 7 \end{bmatrix}$$

17. The linear system $\begin{cases} x+2y+5z=4 \\ 2x+5y+13z=8 \\ 4x+5y+12z=19 \\ 3x+9y+25z=k \end{cases}$ has a unique solution if k is equal to:

Answer: $k=15$; putting the system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 2 & 5 & 13 & 8 \\ 4 & 5 & 12 & 19 \\ 3 & 9 & 25 & k \end{array}\right]$$

and proceeding with Gaussian elimination

$$\xrightarrow[r_4-3r_1 \to r_4]{r_2-2r_1 \to r_2,\ r_3-4r_1 \to r_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 3 & 0 \\ 0 & -3 & -8 & 3 \\ 0 & 3 & 10 & k-12 \end{array}\right]$$

$$\xrightarrow[r_4-3r_2 \to r_4]{r_3+3r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & k-12 \end{array}\right]$$

$$\xrightarrow{r_4-r_3 \to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & k-15 \end{array}\right]$$

then the system is only consistent if

$$\#\text{ leading variables} = \#\text{ non-zero rows} = \#\text{ unknowns} = 3$$

then $k=15$.

18. The roots of $P(x)=x^4+\frac{1}{2}x^2+3x+2=0$ may include:

Answer: one of the possible negative rational roots -1, -2, $-1/2$, -4; firstly, realise that for us to use any of the theorems/rules (except Descartes' rule of signs. That only requires real coefficients) at all, we would need the coefficients to be **integers**. Multiply by 2 and name the new polynomial $Q(x)$.

$$Q(x)=2x^4+x^2+6x+4=0$$

secondly, use Descartes' rule of signs

$$\begin{aligned} P(x) &= 2x^4+x^2+6x+4 && \text{(doesn't change signs, n}(P)=0) \\ P(-x) &= 2x^4+x^2-6x+4 && \text{(changes signs 2 times, n}(N)=\{2,\,0\}) \end{aligned}$$

then, using the rational zero test

$$a_0=4 \Rightarrow \text{factors}(a_0)=\pm 1,\ \pm 4,\ \pm 2$$

$$a_4=2 \Rightarrow \text{factors}(a_4)=\pm 1,\ \pm 2$$

$$\text{factors}(a_0/a_4)=\pm 1,\ \pm 1/2,\ \pm 2,\ \pm 4$$

then, refer back to Descartes' rule of signs: there exist no positive rational roots. So the remaining rational roots (if they exist) must be the negative rational roots. (none of the roots give $P(x) = 0$, but the point is to test your knowledge of Descartes' rule of signs and the rational zero test.)

20. The linear system $\begin{cases} 2x + y + 3z - 8w = 4 \\ 3x + 4y + z + 2w = 3 \\ 12x + 11y + 11z + aw = b \end{cases}$ is consistent when:

Answer: $a = -20$, $b = 18$; putting the system into augmented matrix form

$$\left[\begin{array}{cccc|c} 2 & 1 & 3 & -8 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 12 & 11 & 11 & a & b \end{array}\right]$$

proceeding with Gaussian elimination

$$\xrightarrow{(1/2)r_1 \to r_1} \left[\begin{array}{cccc|c} 1 & 1/2 & 3/2 & -4 & 2 \\ 3 & 4 & 1 & 2 & 3 \\ 12 & 11 & 11 & a & b \end{array}\right]$$

$$\xrightarrow[r_3 - 12r_1 \to r_3]{r_2 - 3r_1 \to r_2} \left[\begin{array}{cccc|c} 1 & 1/2 & 3/2 & -4 & 2 \\ 0 & 5/2 & -7/2 & 14 & -3 \\ 0 & 5 & -7 & 48 + a & b - 24 \end{array}\right]$$

$$\xrightarrow{(2/5)r_2 \to r_2} \left[\begin{array}{cccc|c} 1 & 1/2 & 3/2 & -4 & 2 \\ 0 & 1 & -7/5 & 28/5 & -6/5 \\ 0 & 5 & -7 & 48 + a & b - 24 \end{array}\right]$$

$$\xrightarrow{r_3 - 5r_2 \to r_3} \left[\begin{array}{cccc|c} 1 & 1/2 & 3/2 & -4 & 2 \\ 0 & 1 & -7/5 & 28/5 & -6/5 \\ 0 & 0 & 0 & 20 + a & b - 18 \end{array}\right]$$

then the final row has to be all zeroes; $a = -20$, $b = 18$

2 Multiple methods for P.19

Solution 1

Answer: (choice is missing); if we look closely, the root r cannot be 0. That's due to the fact the constant term 1 exists. And since the root r is of multiplicity 3, $a \neq 0$.

Let us define $q(x)$ as $P(x)/a$

$$q(x) = \frac{P(x)}{a} = x^3 + \frac{1}{a}x^2 + \frac{b}{a}x + \frac{1}{a}$$

and since, $P(r) = 0$, then $q(r) = 0$ and has multiplicity 3

$$\begin{aligned} q(x) = (x - r)^3 &= x^3 - 3x^2 r + 3xr^2 - r^3 \\ &= x^3 + \frac{1}{a}x^2 + \frac{b}{a}x + \frac{1}{a} \end{aligned}$$

matching coefficients

$$-3r = -r^3 = \frac{1}{a} \Rightarrow r(r^2 - 3) = 0 \Rightarrow r = \pm\sqrt{3},\ a = \mp\frac{\sqrt{3}}{9}$$

$$\frac{b}{a} = 3r^2 \Rightarrow b = 3r^2 a = \mp\sqrt{3}$$

so that $P(x) = \pm\frac{\sqrt{3}}{9}(x \pm \sqrt{3})^3$. Generally, r has a sign opposite to that of a or b.

Solution 2

Answer: (choice is missing); notice that if the root is repeated 3 times, then no other roots exist (since this is a third degree equation). Then r must lie on either side of 0 (for the reason mentioned in the previous method, the constant term).

Then, if $r > 0$, the equation (using Descartes' rule of signs) has to yield only positive roots and no negative roots.

$$P(-x) = -ax^3 + x^2 - bx + 1$$

then for $\mathrm{n}(N) = 0$, a and b have to have negative signs (opposite to r).

If $r < 0$, the equation (using Descartes' rule of signs) has to yield only negative roots and no positive roots.

$$P(x) = ax^3 + x^2 + bx + 1$$

then for $\mathrm{n}(P) = 0$, a and b have to have positive signs (opposite to r).